VMC Vidyamandir Classes SINCE 1986

IIT JEE | MEDICAL | FOUNDATION

CODE A

JEE Main - 14 | JEE - 2022

Date: 31/10/2021 **Maximum Marks: 300**

Timing: 04:00 PM to 09:00 PM

General Instructions

1. The test is of **3 hours** duration.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them**.
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) : _

Roll Number : _

OMR Bar Code Number : _

Candidate's Signature : _ _ _ _ _ _ _ _ _ _ **Invigilator's Signature** _ _ _ _ _ _ _ _

PART I : PHYSICS 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. A series combination of a resistance and a capacitor of capacitance $5\ \mu F$ is connected to an AC source of variable frequency. The combination has an impedance $1250\ \Omega$ at angular frequency 200 rad/s. The impedance (in Ohms) of the combination at angular frequency 400 rad/s will be:

(A) 1530.9 **(B)** 1140.2 **(C)** 901.4 **(D)** 689.9

2. An equilateral prism is placed in a medium of air. A ray of light travelling in air is incident on one face of the prism at an angle of incidence θ_I. The ray emerges out of the prism with the minimum net deviation for $\theta_I = 45^o$. The refractive index of the prism is:

(A) $\sqrt{\frac{3}{2}}$ **(B)** $\sqrt{2}$ **(C)** $\sqrt{3}$ **(D)** 2

3. A resistance and an inductor coil are connected in series with an AC voltage source of variable frequency. If, starting from a value f_0, the frequency of the source is increased, then the rate of heat dissipation in the resistance:

(A) increases continuously **(B)** decreases continuously

(C) increases first, then decreases **(D)** may increase or decrease, depending on f_0

4. In a series LCR circuit powered by an AC voltage source $V(t) = 100\cos(200t)$, the current varies as $I(t) = 2\cos\left(200t - \frac{\pi}{3}\right)$, with all quantities being in SI units. The value of the resistance in the circuit, in Ohms, is:

(A) $50\sqrt{3}$ **(B)** 50 **(C)** $25\sqrt{3}$ **(D)** 25

5. In a Young's double slit experiment, the slit separation is 0.1 mm, the distance between the slits and the screen is 4 m, and monochromatic light of wavelength 500 nm is used. The fringe width (in cm) of the interference pattern is:

(A) 0.1 **(B)** 0.2 **(C)** 1.0 **(D)** 2.0

6. A thin equi-convex lens of focal length 40 cm is placed in contact with the reflecting surface of a plane mirror, such that the principle axis of the lens is perpendicular to the surface of the mirror. The distance (in cm) from the pole of the lens at which a small object must be placed, such that a real image 2 times the size of the object is formed, is:

(A) 30 **(B)** 40 **(C)** 60 **(D)** 90

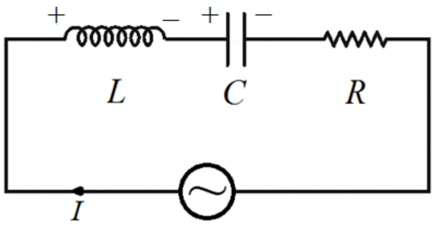
7. The fringe width of the interference pattern in a Young's double slit experiment with identical sized slits is β and the intensity at the central maximum is I_M. The minimum distance between any two closest points on the screen where the intensity is $\frac{3I_M}{4}$ is:

(A) $\frac{2\beta}{3}$ **(B)** $\frac{\beta}{2}$ **(C)** $\frac{\beta}{3}$ **(D)** $\frac{\beta}{4}$

8. An object is placed 60 cm in front of a convex mirror of focal length 20 cm. The image formed is:

(A) real and 3 times the size of the object
(B) virtual and 4 times the size of the object
(C) virtual and 1/3 times the size of the object
(D) virtual and 1/4 times the size of the object

9. In a series LCR circuit powered by an AC voltage source, let the instantaneous potential difference across the capacitor, the instantaneous potential difference across the inductor, and instantaneous current in the circuit be denoted by V_C, V_L and I respectively, with the polarities of these quantities being those shown in the figure. All quantities being in SI units, which of these combinations of V_C, V_L and I is possible?



(A) $V_C = 200\sin\left(300t - \frac{\pi}{4}\right), V_L = 200\sin\left(300t + \frac{3\pi}{4}\right), I = 3\sin\left(300t - \frac{\pi}{4}\right)$

(B) $V_C = 200\sin\left(300t - \frac{\pi}{4}\right), V_L = 400\sin\left(300t + \frac{3\pi}{4}\right), I = 4\sin\left(300t + \frac{\pi}{4}\right)$

(C) $V_C = 200\sin\left(300t + \frac{2\pi}{3}\right), V_L = 200\sin\left(300t - \frac{\pi}{3}\right), I = 3\sin(300t)$

(D) $V_C = 200\sin\left(300t - \frac{\pi}{3}\right), V_L = 400\sin\left(300t + \frac{\pi}{6}\right), I = 4\sin\left(300t - \frac{\pi}{6}\right)$

10. A ray of light is incident normally on the face AB of a right-angled prism ABC with $\angle A = 30^o$ and $\angle B = 90^o$. The refractive index of the prism is μ and the surrounding medium is air (refractive index 1). Consider the following statements:

(I) If $\mu = 1.5$, the ray emerges into air from the face BC
(II) If $\mu = \sqrt{3}$, the ray suffers a net deviation of 30^o due to the prism

Which of these options is correct?

(A) Both (I) and (II) are correct
(B) (I) is correct and (II) is incorrect
(C) (II) is correct and (I) is incorrect
(D) Both (I) and (II) are incorrect

11. The two refracting surfaces of biconcave lens have radius of curvature 30 cm and 60 cm. The refractive index of the lens is 1.5. The focal length (in cm) of the lens when it is placed in water of refractive index 4/3 is:

(A) 160 **(B)** 133.3 **(C)** 120 **(D)** 91.7

12. A ray of light incident on a plane mirror experiences a deviation of 60^o. The angle of incidence (in degrees) is:

(A) 45 **(B)** 30 **(C)** 60 **(D)** 75

13. In a Young's double slit experiment, the distance between the slits is 0.2 mm, the slits are at a distance 2 m from the screen, and monochromatic light of wavelength 500 nm is used. The distance (in cm) between the central maximum and the 3rd closest minimum to the central maximum is:

(A) 1.00 **(B)** 1.25 **(C)** 1.50 **(D)** 1.75

14. Light from a small, distant object falls on a glass sphere of radius R and refractive index μ placed in a medium of air (refractive index 1). The light is focused by the sphere to a point P. The distance of the point P from the centre of the sphere is:

(A) $\frac{(\mu-1)}{(2-\mu)}R$ **(B)** $\frac{\mu}{(\mu-1)}R$ **(C)** $\frac{(2-\mu)}{2(\mu-1)}R$ **(D)** $\frac{\mu}{2(\mu-1)}R$

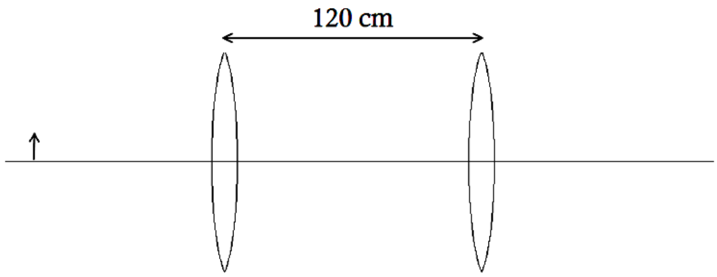
15. In a Young's double slit experiment, the distance between the slits is 0.5 mm. and monochromatic light is used. The wavelength of the light (in nm), for which the angular fringe width in the interference pattern is 1.2×10^{-3} rad , is:

(A) 240 **(B)** 450 **(C)** 600 **(D)** 750

16. In a Young's double slit experiment, the intensity leaving the slits is I and αI respectively, where $0<\alpha<1$. The ratio of the resultant intensity at a maximum on the screen to the resultant intensity at a minimum on the screen is:

(A) $\frac{1+\alpha}{1-\alpha}$ **(B)** $\left(\frac{1+\sqrt{\alpha}}{1-\sqrt{\alpha}}\right)^2$ **(C)** $\frac{1+\sqrt{\alpha}}{1-\sqrt{\alpha}}$ **(D)** $\frac{2+\alpha}{2-\alpha}$

17. Two convex lenses, both of focal length 45 cm, are placed with their axes coincident and a distance 120 cm between them. An object is placed on their common axis. The distance (in cm) of the object from one of the lenses such that after refraction by both lenses, a real image the same size as the object is formed, is:



(A) 240 **(B)** 180 **(C)** 120 **(D)** 75

18. For a series LCR circuit connected to an AC voltage source operating at frequency f , the power factor is:
(Assume that the voltage amplitude of the source is independent of the frequency)

(I) the ratio of the resistance to the impedance of the circuit

(II) the square root of the ratio of the power consumed at frequency f to the power consumed at the resonant frequency

Which of these options is correct?

(A) Both (I) and (II) are correct **(B)** (I) is correct and (II) is incorrect

(C) (II) is correct and (I) is incorrect **(D)** Both (I) and (II) are incorrect

19. A plano-convex lens of radius of curvature R_1 and refractive index μ_1 is placed in close contact with a biconcave lens of radius of curvature R_2 and refractive index μ_2. The combination is placed in air of refractive index 1. The combination behaves like a converging lens if:

(A) $\frac{R_1}{(\mu_1-1)}<\frac{R_2}{2(\mu_2-1)}$ **(B)** $\frac{R_1}{(\mu_1-1)}<\frac{2R_2}{(\mu_2-1)}$

(C) $\frac{R_1}{(\mu_1-1)}>\frac{R_2}{2(\mu_2-1)}$ **(D)** $\frac{R_1}{(\mu_1-1)}>\frac{2R_2}{(\mu_2-1)}$

20. A small object is placed on the principal axis of a convex lens of focal length *f*, at a distance $2f+\delta$ from the optical centre of the lens, such that $\delta \ll 2f$. The magnification produced by the lens is:

(A) $1-\dfrac{\delta}{2f}$ **(B)** $1-\dfrac{\delta}{f}$ **(C)** $1+\dfrac{\delta}{2f}$ **(D)** $1+\dfrac{\delta}{f}$

SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. ***In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*****. (Example: 6, 81, 1.50, 3.25, 0.08)**

21. The quality factor of a series circuit consisting of a resistance $5\ \text{k}\Omega$, a coil of inductance 20 mH, a capacitor of capacitance $8\ \mu\text{F}$ and an AC voltage source is __________.

22. In a Young's double slit experiment, the distance between the slits is 5 mm, the slits are at a distance 1 m from the screen, and monochromatic light of wavelength 600 nm is used. If a thin transparent sheet of thickness 0.2 mm and refractive index 1.5 is introduced in front of one of the slits, the interference pattern shifts on the screen by a distance __________ cm.

23. A resistance $6\ \Omega$ and a coil of inductance 40 mH are connected to an AC voltage source of angular frequency 200 rad/s. The rms current in the circuit is 3 A. A capacitor is now connected in series with the resistance and the coil. The rms current in the circuit is now 4 A, and now the current leads the voltage source. The capacitance $(\text{in } \mu\text{F})$ of the capacitor is _________.

24. A concave mirror is fixed on the ground with its principle axis vertical and its reflecting surface facing upwards, and filled with water (refractive index 4/3). A small object is placed on the axis of the mirror at a distance 120 cm from it. The final image of the object is coincident with it. The focal length (in cm) of the mirror is __________.

25. In a Young's double slit experiment performed in a medium of air with monochromatic light; the fringe width is 6 mm. The apparatus is now completely submerged inside water (refractive index $\dfrac{4}{3}$) and the experiment is performed again, with the distance between the slits and the screen being 2 times that in the first experiment. The source of the light used and the slit separation are still the same as in the first experiment. The fringe width in the second experiment is ____________mm.

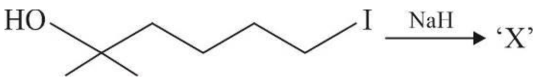
SPACE FOR ROUGH WORK

PART II : CHEMISTRY 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Consider the reaction Scheme-I and II. Identify correct statement regarding organic products X and Y.

I. 

II. 

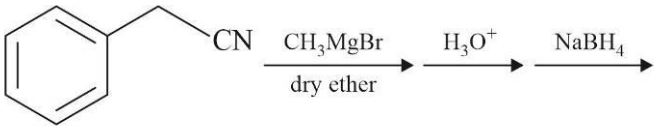
(A) Product 'X' and 'Y' are identical

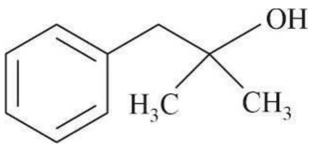
(B) Product 'X' and 'Y' are stereoisomers

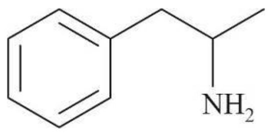
(C) Product 'Y' form instant turbidity with Lucas reagent

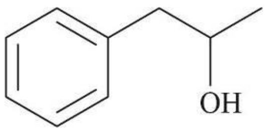
(D) Product 'Y' on reaction with H^+ as catalyst form product 'X'

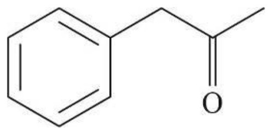
2. Identify major product of the following reaction sequence.



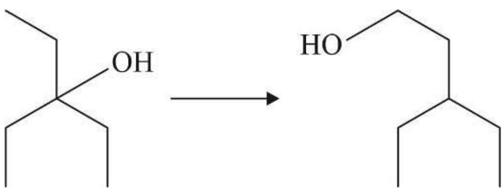
(A) 

(B) 

(C) 

(D) 

3. Identify correct sequence of reagents used for the following conversion.



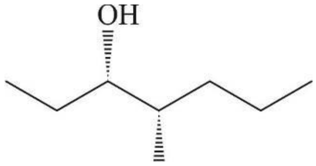
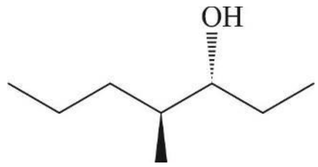
(A) H_2SO_4 / Δ, NBS, NaOH(aq), H_2 / Pd

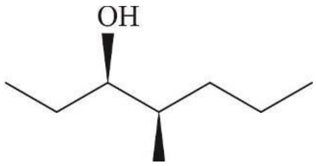
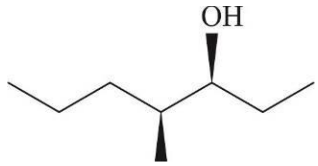
(B) H_2SO_4 / Δ, HBr / peroxide, MeOK / Δ, $B_2H_6 / H_2O_2 - OH^-$

(C) H_2SO_4 / Δ, HBr / peroxide, t-BuOK / Δ, $B_2H_6 / H_2O_2 - OH^-$

(D) Red hot Cu, $NaNH_2$, $B_2H_6 / H_2O_2 - OH^-$

4. Pheromones are chemicals used by insects and some animals to communicate. For example the ant species Leptogenysdiminuta secretes one of the isomer of 4-methyl-3-heptanol to mark trails to food. The active stereoisomer is (3R-4S)-4-methyl-3-heptanol, which of these stereoisomers is (3R-4S)-4-methyl-3-heptanol ?

(A) 
(B) 

(C) 
(D) 

5. Menthol is 2-isopropyl-5-methylcyclo-hexanol. How many stereoisomers of menthol have the isopropyl group trans to the hydroxyl group ?

(A) 2 **(B)** 4 **(C)** 6 **(D)** 8

6. Aqueous solution having complex ion $[Cu(H_2O)_4]^{2+}$ is blue in colour. What is the colour of the light absorbed by the complex.

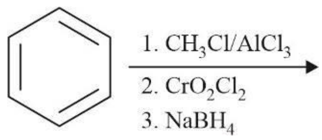
(A) Yellow **(B)** Blue **(C)** Red **(D)** Green

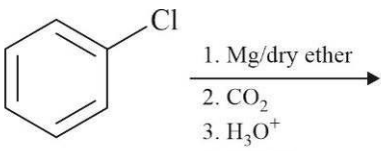
7. Which of the following is correct IUPAC name of the Wilkinson's catalyst ?

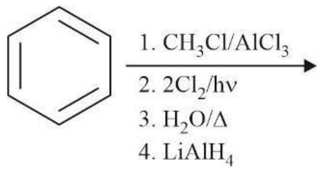
(A) Dichloridobis(triphenylphosphine)nickel(II)
(B) Chloridotris(triphenylphosphine)rhodium(I)
(C) cis-dichloridodiammineplatinum(II)
(D) Dichloridobis(triphenylphosphine)rhodium(II)

8. Which of the following reaction schemes produce benzyl alcohol ?

I. $HC \equiv CH \xrightarrow[\substack{2.\ Cl_2/Fe \\ 3.\ Mg/dry\ ether \\ 4.\ CH_2O \\ 5.\ H_3O^+}]{1.\ Red\ hot\ Fe\text{-}tube}$

II. 

III. 

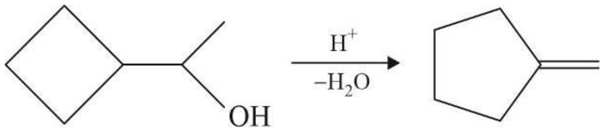
IV. 

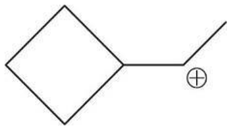
(A) II, III, IV **(B)** I, II, III, IV **(C)** I, II, IV **(D)** I, III, IV

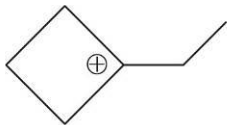
9. Which of the following is not correct for the complex ion $[Co(C_2O_4)_3]^{3-}$?

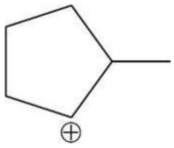
(A) It is an inner orbital complex
(B) It is para magnetic complex
(C) It exist in two stereoisomeric form
(D) It has twelve 90° $O-Co-O$ bond angles

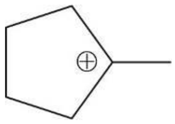
10. Which of the following carbocation will not be formed as an intermediate in the following reaction ?



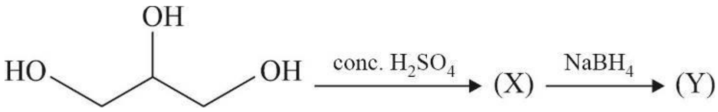
(A) 

(B) 

(C) 

(D) 

11. Consider the following reaction sequence.

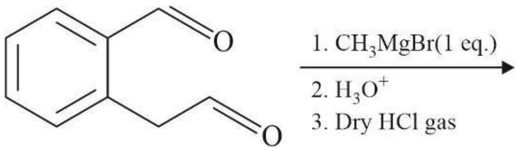


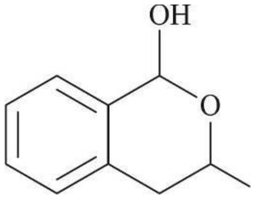
$(Y) \xrightarrow{(Z)} (X)$

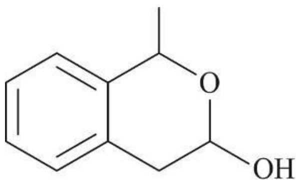
Which of the following is not correct for (Z) ?

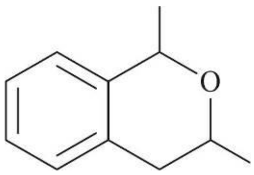
(A) Red hot Cu **(B)** PCC **(C)** MnO_2 **(D)** $K_2Cr_2O_7 / H^+$

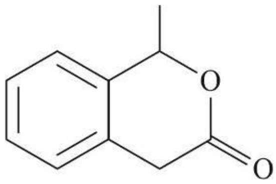
12. Identify major product of the following reaction.



(A) 

(B) 

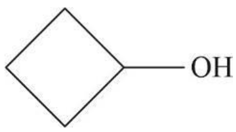
(C) 

(D) 

13. One of the isomer of tetra hydro furan on reaction with B_2H_6 form H_2 gas. Identify this isomer.

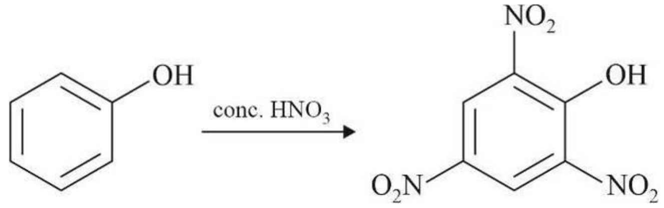
(A) $CH_3-\underset{\underset{O}{\|}}{C}-CH_2-CH_3$

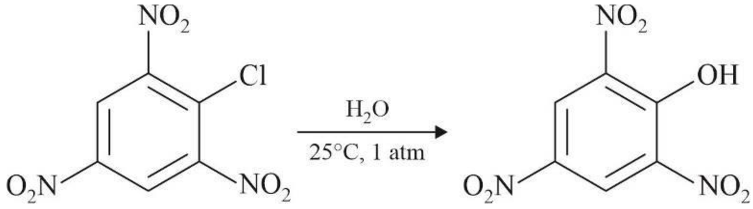
(B) $CH_2=CH-CH_2-OCH_3$

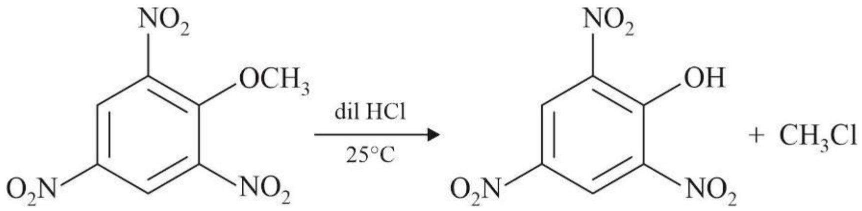
(C) 

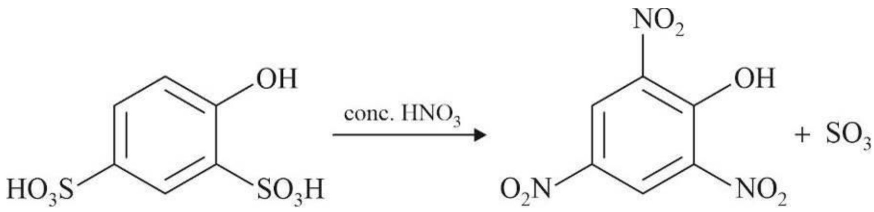
(D) $CH_3CH_2CH_2CH_2OH$

14. Which of the following reaction produce high yield of the picric acid ?

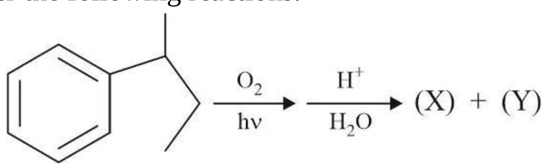
(A) 

(B) 

(C) 

(D) 

15. Consider the following reactions.



In the presence of air, (X) is slowly oxidized to dark coloured mixtures containing quinones. Identify incorrect statement.

(A) (X) form white precipitate with Br_2 water

(B) (Y) form yellow precipitate with I_2 and NaOH

(C) (X) gives dark pink colour with NaOH solution

(D) (X) gives positive neutral $FeCl_3$ test

16. Which of the following is commercial method for production of methanol ?

(A) $CH_2=O \xrightarrow{NaBH_4} CH_3OH$

(B) $2CH_4+O_2 \xrightarrow[523K,\ 100atm]{Cu} 2CH_3OH$

(C) $CH_3Cl+NaOH \longrightarrow CH_3OH+NaCl$

(D) $CO+2H_2 \xrightarrow[573K,\ 200atm]{ZnO,\ Cr_2O_3} CH_3OH$

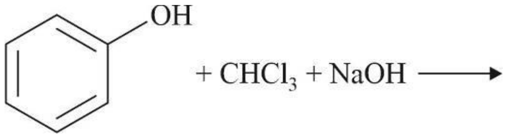
17. The conductance measurement indicate two ions per formula unit for two isomeric octahedral coordination compounds (X) and (Y) having formula $Co(C_2O_4)Br \cdot 4NH_3$.

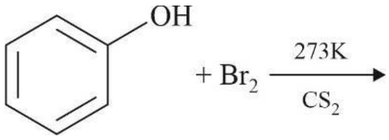
Coordination compound (X) and (Y) do not produce white precipitate of calcium oxalate on reaction with $Ca(NO_3)_2$. Compound (X) can give pale yellow precipitate with $AgNO_3$ solution, while compound (Y) can't show $AgNO_3$ test. Identify correct statement :

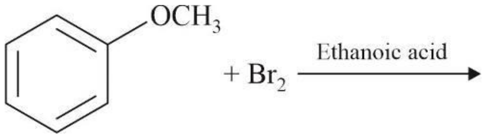
I. (X) has coordination sphere as cation
II. (Y) has coordination sphere as anion
III. (Y) has several stereoisomers
IV. Only one compound is possible

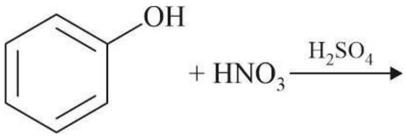
(A) I, IV only **(B)** II, III only **(C)** I, II, III only **(D)** None of these

18. In which of the following reaction major product is an ortho-isomer ?

(A) 

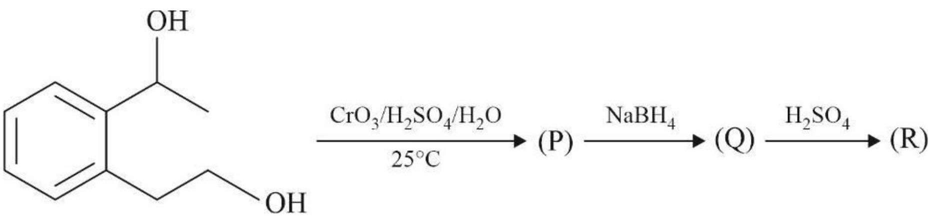
(B) 

(C) 

(D) 

19. Which of the following is correct statement ?

(A) $[Ni(CO)_4]$ shows synergic bonding
(B) $[Ni(CO)_4]$ is square planar
(C) $[Ni(CO)_4]$ is paramagnetic
(D) $[Ni(CO)_4]$ has coordination number 8

20. 

Consider above reaction sequence. Identify correct statement.

(A) (P) is dicarboxylic acid
(B) (P) is keto aldehyde
(C) (Q) is diol
(D) (R) is cyclic ester

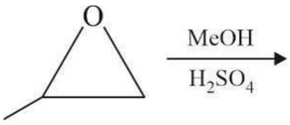
SPACE FOR ROUGH WORK

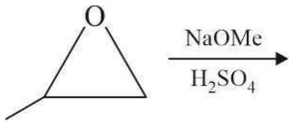
SECTION-2

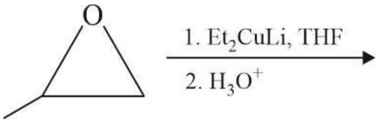
This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. ***In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*****. (Example: 6, 81, 1.50, 3.25, 0.08)**

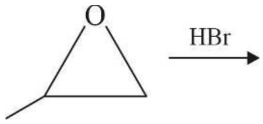
21. Cerric ammonium nitrate is used as analytical reagent for identification of alcoholic functional group in an organic compound. What is sum of coordination number and effective atomic number of the central atom in cerric ammonium nitrate ? [Ce = 58]

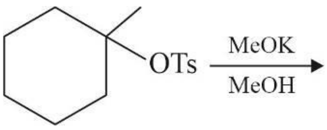
22. In how many of the following reactions neither oxidation nor reduction happens to the organic molecule during each reaction.

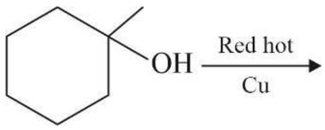
(a) 

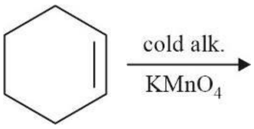
(b) 

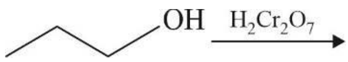
(c) 

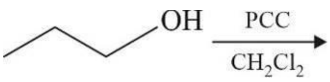
(d) 

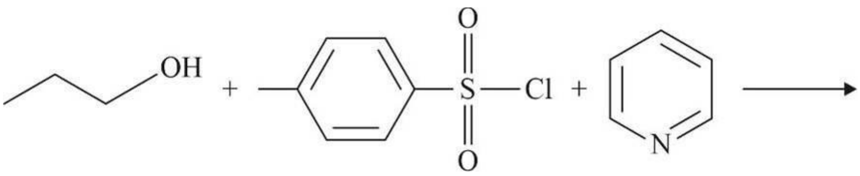
(e) 

(f) 

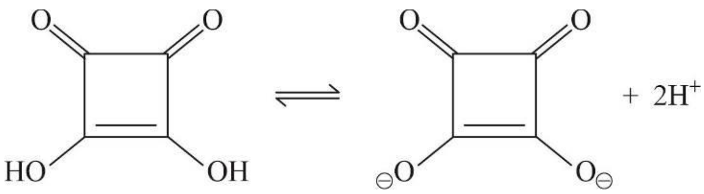
(g) 

(h) 

(i) 

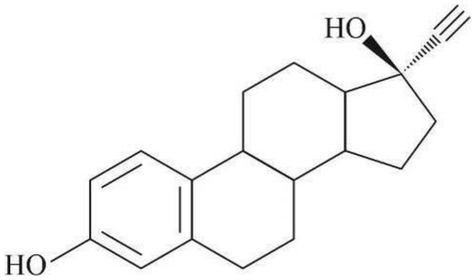
(j) 

23. Squaric acid shown below, can lose two protons to become the squarate dianion.



The average bond order of each $C-O$ bond of squarate dianion is X and it is aromatic having $(4n+2)\pi$ electrons. What is value of $(2X+n)$?

24. The structure of ethyl estradiol, an oral contraceptive, is shown below.



This molecule contains X_1 chiral carbon atoms, X_2 saturated tertiary carbon atoms and X_3 degree of unsaturation. What is the numerical value of $(X_1 + X_2 + X_3)$.

25. Consider the structure of complex compound $[Fe(EDTA)]^{2-}$ where EDTA is ethylenediamine tetraacetate ligand. This complex contains Z_1 rings, Z_2 non-coordinated lone pair electrons of ligand and Z_3 is number of 90° $N-Fe-O$ bond angle. Find the numerical value of $(Z_1 + Z_2 + Z_3)$.

SPACE FOR ROUGH WORK

PART III : MATHEMATICS 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. If $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are non-zero vectors satisfying $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d})-(\vec{b}\times\vec{c})\times(\vec{d}\times\vec{a})=[\vec{a}\ \vec{c}\ \vec{d}]\vec{b}$ then :

(A) no three out of $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are coplanar

(B) $\vec{b}, \vec{c}, \vec{d}$ are coplanar

(C) $\vec{a}, \vec{b}, \vec{d}$ are coplanar

(D) $\vec{a}, \vec{b}, \vec{c}$ are coplanar

2. Equation of the line in the plane $P: x+3y-z=9,$ which is perpendicular to the line $\vec{r}=\hat{i}+\hat{j}+\hat{k}+\lambda(2\hat{i}+\hat{j}-\hat{k})$ and passing through a point where the plane P meets the given line, is :

(A) $\frac{x-3}{2}=\frac{y-2}{1}=\frac{z}{5}$ **(B)** $\frac{x-3}{-2}=\frac{y-2}{-1}=\frac{z}{5}$

(C) $\frac{x-3}{-5}=\frac{y-2}{1}=\frac{z}{2}$ **(D)** $\frac{x-3}{1}=\frac{y-2}{5}=\frac{z}{2}$

3. The equation of the right bisector plane of the segment joining (2, 3, 4) and (6,7, 8) is :

(A) $x+y+z+15=0$ **(B)** $x+y+z-15=0$

(C) $x-y+z-15=0$ **(D)** None of these

4. Consider the lines $L_1; \vec{r}=(\lambda+1)(\hat{i}+\hat{j}+\hat{k})$ and $L_2: \vec{r}\cdot(2\hat{i}+\hat{j}+\hat{k})-1=0=\vec{r}\cdot(3\hat{i}+\hat{j}+2\hat{k})-2,$ then shortest distance between them is :

(A) $\sqrt{2}$ **(B)** $2\sqrt{2}$ **(C)** $\frac{1}{\sqrt{2}}$ **(D)** None of these

5. Equation of a line passing through point $P(3,-4,1)$ and meeting line $\frac{x-3}{2}=\frac{y+1}{-3}=\frac{z-2}{-1}$ at point $Q,$ such that PQ is parallel to plane $2x+y-z=5$ is :

(A) $\frac{x-3}{1}=\frac{y+4}{-3}=\frac{z-1}{-1}$ **(B)** $\frac{x-3}{1}=\frac{y+4}{3}=\frac{z+1}{-1}$

(C) $\frac{x-3}{1}=\frac{y-4}{-3}=\frac{z-1}{1}$ **(D)** $\frac{x+3}{1}=\frac{y+4}{-3}=\frac{z-1}{-1}$

6. Distance of origin from plane containing lines $L_1: \vec{r}=(2+\lambda)\hat{i}+(1+7\lambda)\hat{j}+(-2-5\lambda)\hat{k}$ and $L_2: \vec{r}=(4+\mu)\hat{i}+(-3+\mu)\hat{j}-\mu\hat{k}$ is :

(A) $\frac{1}{\sqrt{14}}$ **(B)** $\frac{2}{\sqrt{14}}$ **(C)** $\frac{3}{\sqrt{14}}$ **(D)** $\frac{4}{\sqrt{14}}$

7. Let $L_1 : \vec{r} = \hat{i} - \hat{j} - 10\hat{k} + \lambda(2\hat{i} - 3\hat{j} + 8\hat{k})$ and $L_2 : \vec{r} = 4\hat{i} - 3\hat{j} - \hat{k} + \mu(\hat{i} - 4\hat{j} + 7\hat{k})$ represent two lines in R^3, then which one of the following is incorrect?

(A) L_1 is parallel to the vector $4\hat{i} - 6\hat{j} + 16\hat{k}$

(B) L_2 is parallel to the vector $-\hat{i} + 4\hat{j} - 7\hat{k}$

(C) L_1 and L_2 are coplanar

(D) Angle between the lines L_1 and L_2 is $\cos^{-1}\left(\frac{70}{11\sqrt{7}}\right)$

8. The equation of the line passing through $M(1, 1, 1)$ and intersects at right angle to the line of intersection of the planes $x + 2y - 4z = 0$ and $2x - y + 2z = 0$ is $\frac{x-1}{a} = \frac{y-1}{b} = \frac{z-1}{c}$, then $a : b : c$ equals :

(A) $5 : -1 : 2$ **(B)** $-5 : 1 : 2$ **(C)** $5 : -1 : -2$ **(D)** $5 : 1 : 2$

9. The shortest distance between z axis and the line $\frac{x-2}{3} = \frac{y-5}{2} = \frac{z+1}{-5}$ is equal to :

(A) $\frac{11}{\sqrt{13}}$ **(B)** $\frac{17}{\sqrt{13}}$ **(C)** $\frac{11}{13}$ **(D)** $\frac{\sqrt{11}}{13}$

10. The distance of the point $(1, -5, 9)$ from the plane $x - y + z = 5$ measured along the line $x = y = z$ is :

(A) $\frac{20}{3}$ **(B)** $3\sqrt{10}$ **(C)** $10\sqrt{3}$ **(D)** $\frac{10}{\sqrt{3}}$

11. The value of $|(\vec{r} \cdot \vec{a})(\vec{b} \times \vec{c}) + (\vec{r} \cdot \vec{b})(\vec{c} \times \vec{a}) + (\vec{r} \cdot \vec{c})(\vec{a} \times \vec{b})|$, where $|\vec{r}| = 2$ and $\vec{a}, \vec{b}, \vec{c}$ are coterminous sides of a tetrahedron with volume 3, is equal to :

(A) 6 **(B)** 12 **(C)** 18 **(D)** 36

12. The value of `a' for which the straight line $4x - 3y + 4z - 2 = 0 = 3x - 2y + z - 5$ is parallel to the plane $2x - y + az - 7 = 0$, will be :

(A) 8 **(B)** –8 **(C)** 11 **(D)** –2

13. The ratio in which the plane $\vec{r} \cdot (\hat{i} - 2\hat{j} + 3\hat{k}) = 17$ divides the line joining the points $-2\hat{i} + 4\hat{j} + 7\hat{k}$ and $3\hat{i} - 5\hat{j} + 8\hat{k}$ is given by :

(A) $1 : 5$ **(B)** $1 : 10$ **(C)** $3 : 5$ **(D)** $3 : 10$

14. Consider a variable plane $lx + my + nz = k(k > 0)$ and l, m, n are direction cosines of normal of the plane. Let the given plane intersects the co-ordinate axes at A, B and C, then the minimum area of ΔABC is :

(A) $\frac{3\sqrt{3}k^2}{2}$ **(B)** $\frac{3\sqrt{3}\,k^2}{4}$ **(C)** $3\sqrt{3}\,k^2$ **(D)** $12\sqrt{3}\,k^2$

15. If the plane passing through the points $(a, 1, 1), (1, 2, 1)$ and $(2, 3, 4)$ is parallel to the line $\vec{r} = \lambda(\hat{i} + \hat{j} + 2\hat{k})\ (\lambda \in R)$, then a is equal to :

(A) $-\frac{1}{2}$ **(B)** -1 **(C)** $\frac{3}{2}$ **(D)** 0

16. Let $\vec{r} = (\vec{a}\times\vec{b})\sin x + (\vec{b}\times\vec{c})\cos y + 2(\vec{c}\times\vec{a})$ where $\vec{a}, \vec{b}, \vec{c}$ are three noncoplanar vectors. If $\vec{r}$ is perpendicular to $\vec{a}+\vec{b}+\vec{c}$, then minimum value of $x^2 + y^2$ is :

(A) π^2 **(B)** $\frac{\pi^2}{4}$ **(C)** $\frac{5\pi^2}{4}$ **(D)** None of these

17. If $\vec{a} = \vec{b}+\vec{c}, \vec{b}\times\vec{d} = \vec{0}$ and $\vec{c}\cdot\vec{d} = 0$, then $\frac{\vec{d}\times(\vec{a}\times\vec{d})}{\vec{d}^2}$ is equal to :

(A) $\vec{a}$ **(B)** $\vec{b}$ **(C)** $\vec{c}$ **(D)** $\vec{d}$

18. The equation of the acute angle bisector of planes $2x - y + z - 2 = 0$ and $x + 2y - z - 3 = 0$ is :

(A) $x - 3y + 2z + 1 = 0$ **(B)** $3x + 3y - 2z + 1 = 0$

(C) $x + 3y - 2z + 1 = 0$ **(D)** $3x + y = 5$

19. If $|\vec{a}| = 2, |\vec{b}| = 5$ and $\vec{a}\cdot\vec{b} = 0$, then $\vec{a}\times(\vec{a}\times(\vec{a}\times(\vec{a}\times(\vec{a}\times(\vec{a}\times\vec{b})))))$ is equal to :

(A) $64\vec{a}$ **(B)** $64\vec{b}$ **(C)** $-64\vec{b}$ **(D)** $-64\vec{a}$

20. Find the equation of the plane through (3, 4,1) which is parallel to the plane $\vec{r}\cdot(2\vec{i} - 3\vec{j} + 5\vec{k}) + 7 = 0$.

(A) $\vec{r}\cdot(2\vec{i} - 3\vec{j} + 5\vec{k}) + 1 = 0$ **(B)** $\vec{r}\cdot(2\vec{i} + 3\vec{j} + 5\vec{k}) - 1 = 0$

(C) $\vec{r}\cdot(2\vec{i} + 3\vec{j} + 5\vec{k}) + 1 = 0$ **(D)** $\vec{r}\cdot(2\vec{i} - 3\vec{j} + 5\vec{k}) - 1 = 0$

SPACE FOR ROUGH WORK

SECTION-2

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21. Let $P(a, b, c)$ be any point on the plane $3x+2y+z=7$, then the least value of $(a^2+b^2+c^2)$, is :

22. If $\hat{a}$ and $\hat{b}$ are unit vectors such that $[\hat{a}\ \hat{b}\ \hat{a}\times\hat{b}]=\frac{1}{4}$ and acute angle between $\hat{a}$ and $\hat{b}$ is $\frac{\pi}{k}$, then k equals :

23. If distance between two non-intersecting planes P_1 and P_2 is 3 units, where P_1 is $2x-3y+6z+5=0$ and P_2 is $4x+by+cz+d=0$ and point $A(-3, 0, -1)$ is lying between the planes P_1 and P_2 then the value of $(b+c+d)$, is equal to :

24. If $\vec{a}=2\hat{i}+\hat{j}-2\hat{k}, \vec{b}=\hat{i}+\hat{j}, \vec{c}$ are vectors such that $\vec{a}\cdot\vec{c}=|\vec{c}|, |\vec{c}-\vec{a}|=2\sqrt{2}$ and the angle between $\vec{a}\times\vec{b}$ and $\vec{c}$ is 30°, then the value of $10|(\vec{a}\times\vec{b})\times\vec{c}|$ is :

25. Let $\vec{r}$ be a position vector of a variable point in x-y plane such that $\vec{r}\cdot(\vec{r}+8\hat{i}-10\hat{j})+41=0$, then minimum value of $|\vec{r}+2\hat{i}-3\hat{j}|^2$ is :

SPACE FOR ROUGH WORK

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